

Northwind Multi-Table Sales Analysis

Task 25 • Data Analytics Track • SQL + Power BI • Today's Work Report

1. Assigned Task

The assigned task was to combine orders, products, and customers for business analysis and apply relational analysis end-to-end. The required tools were SQL and Power BI. Deliverables were an SQL script, a dashboard, and five business insights. The task guidance specifically highlighted validating joins and avoiding double counting.

2. Work Completed

- Used the Northwind dataset for multi-table sales analysis.
- Prepared a combined analysis dataset containing customer, order, product, quantity, pricing, discount, and sales fields.
- Used SQL-style relational analysis to bring order, customer, and product information together for reporting.
- Built a Power BI dashboard named 'Northwind Sales Analysis' using KPI cards, bar charts, a monthly trend line chart, a detail table, and interactive filters.
- Added filters for Order Date, Order Day, Order Month, Customer Name, and Product Name.

3. Dataset Summary

Metric	Result
Rows in analysis dataset	58
Distinct orders	40
Unique customers	15
Unique products	24
Total quantity	2,942
Total sales	68,137.00
Date range	15 Jan 2006 to 23 Jun 2006

4. SQL Analysis Explanation

The SQL part of the task focused on relational analysis: connecting customer information with orders and product information, selecting the fields needed for sales reporting, and calculating sales-related measures. The resulting dataset contains customer_name, order_id, order_date, product_name, quantity, unit_price, discount, and sales_amount.

Join validation: The output contains 58 line-level records but only 40 distinct order IDs. This is important because one order can contain multiple product lines. Counting rows as orders would therefore overstate the number of orders.

5. Power BI Dashboard Explanation

The Power BI report converts the SQL/data-preparation output into an interactive business dashboard. It contains KPI cards for sales, orders, and quantity; a Top 5 Products by Sales bar chart; a Top 5 Customers by Sales bar chart; a Monthly Sales Trend line chart; a customer sales table; and a dedicated filter panel.

Important validation point: The screenshot currently shows Total Orders as 58. Because the dataset has 58 rows but 40 distinct order_id values, the order KPI should use a distinct count of order_id if the business definition is 'number of orders'. This directly addresses the task hint to avoid double counting.

6. Five Business Insights

Insight 1: Total sales are 68,137 across 40 distinct orders and 2,942 units.

Insight 2: Northwind Traders Coffee is the highest-sales product at 29,900.00, followed by Northwind Traders Beer at 6,818.00.

Insight 3: Company BB is the highest-sales customer at 15,432.50, followed by Company G at 13,800.00.

Insight 4: March has the highest monthly sales at 32,609.25; April is next at 19,355.25.

Insight 5: The analysis covers 15 customers and 24 products from 15 Jan 2006 through 23 Jun 2006; row-level analysis should be distinguished from distinct-order analysis to prevent double counting.

7. Deliverables Completed

- ✓ SQL-based multi-table sales analysis
- ✓ Northwind Sales Analysis Power BI dashboard
- ✓ Five business insights
- ✓ Interactive filtering and dashboard formatting

8. Skills Demonstrated

SQL joins and relational analysis • Aggregation and sales calculations • Join validation • Avoiding double counting • Power BI dashboard design • KPI reporting • Data visualization • Interactive slicers • Business insight generation

9. Conclusion

Today's work completed the main requirements of the Multi-Table Sales Analysis task by combining relational sales data in SQL and presenting the results through an interactive Power BI dashboard. The key validation lesson from the dataset is that line-level rows and distinct orders are different measures; using `DISTINCT order_id` for the order KPI avoids double counting.